down a coastline. The world's largest ice shelves are the Ross Ice Shelf and the Filchner-Ronne ice shelf in Antarctica. **An iceberg** is ice floating in open water that has broken off from glaciers or ice shelf.



Carbon is removed from the atmospheric cycle by cryosphere during the formation of ice and is released when the ice melts.

5.5 Oceans and Seas

The water in the oceans and seas is termed as marine water. Continuous water body that surrounds the continents, created by earth's internal force is known as Ocean. The term ocean takes its origin from the Greek word '***Oceaonus***' meaning enormous river encircling the earth. The area of the World Ocean is 361 million square kilometre. The earth has at present five major oceans: The Pacific Ocean, the Atlantic Ocean, the Indian Ocean, the Arctic Ocean, and the Southern ocean (Figure 5.3). All these oceans are interconnected to form one **Global Ocean or World Ocean**. This nature of water to level up quickly has made it as a reference point to measure the height of the land features and the depth of the sea features.

Fact File

Mean Sea Level (MSL) is the average height of the surface of the sea for all stages of the tide. MSL is reference point to measure the height of land features and depth of the sea features.

Sea is a body of saline water (generally a division of the world ocean) partly or fully enclosed by land. **Marginal sea** is a sea partially enclosed by islands, archipelagos, or peninsulas and extension of oceans towards land. They are generally shallow. Andaman Sea, Arabian Sea, Bay of Bengal, Java Sea, Persian Gulf and Red Sea are marginal seas of the Indian Ocean.

Bay is a water body surrounded on three sides by land and the fourth side (mouth) wide open towards an ocean.

Gulf is a large body of water, with a narrow mouth, that is almost completely surrounded by land. The world's largest gulf is the Gulf of Mexico. **Sound, creek, bight** and **cove** are bays which vary in size and depth.

Strait is a narrow channel of water, connecting two larger bodies of water. Palk Strait connects Gulf of Mannar and Bay of Bengal. **Isthmus** is a narrow strip of land connecting two larger land masses. Isthmus of Suez connects Africa and Asia.

Enclosed seas are seas that reach very deep into the continent stay connected with one or the other ocean of the world through straits. Mediterranean Sea is the best example for enclosed sea. **Partly Enclosed Seas** are those types of seas that are connected to the oceans by a very wide opening and have similar characters of the adjacent ocean. A series of islands may also occur between a partly enclosed sea and the ocean to which it is connected. Caribbean Sea is a perfect example.

Landlocked Seas are completely surrounded by landmass on all sides without any natural outlet. They are actually hyper saline lakes. Dead Sea

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